

Student Performance Dashboard in



Power BI

End-to-End Data Cleaning Pipeline, DAX Measure Architecture & Interactive Dashboard Implementation

BY

Sanket Jingare

Saloni Khillare

Dhananjay Patil

Data Cleaning Overview

 **Multi-Table Ingestion:** Imported 3 relational sources — Student, Marks, and Attendance tables.

 **Primary Key Alignment:** Utilized Student ID across all schemas to build reliable relationships.

Validation & Standardization: Applied standard Power Query transformations to ensure data integrity.

 **Data Model Merging:** Consolidated tables into a unified analytical dataset ready for DAX measures.

	A	B	C	D	E
1	Student ID	Name	Department	Gender	Semester
2	STU001	Aarav Deshmukh	Computer Science	Male	1
3	STU002	Vivaan Shah	Information Technology	Female	2
4	STU003	Aditya Gupta	Electronics	Male	3
5	STU004	Arjun Rao	Mechanical	Female	4
6	STU005	Sai Pawar	Civil	Male	5
7	STU006	Reyansh Chavan	Computer Science	Female	6
8	STU007	Atharv Patil	Information Technology	Male	1
9	STU008	Krishna Kulkarni	Electronics	Female	2
10	STU009	Ishaan Mehta	Mechanical	Male	3
11	STU010	Rohan Singh	Civil	Female	4
12	STU011	Rahul Nair	Computer Science	Male	5
13	STU012	Karan Jadhav	Information Technology	Female	6
14	STU013	Akash Bhosale	Electronics	Male	1
15	STU014	Vikram Jingare	Mechanical	Female	2

	A	B	C	D
1	Student ID	Subject	Mark	Maximum Mark
2	STU001	Mathematics	73	100
3	STU001	Computer Science	70	100
4	STU001	English	93	100
5	STU002	Mathematics	54	100
6	STU002	Computer Science	54	100
7	STU002	English	51	100
8	STU003	Mathematics	57	100
9	STU003	Computer Science	59	100
10	STU003	English	53	100

	A	B
1	Student ID	Attendance %
2	STU001	64.7
3	STU002	69.9
4	STU003	71.8
5	STU004	96.5
6	STU005	94.6
7	STU006	94.5
8	STU007	75.7
9	STU008	67.8
10	STU009	92.8

Power Query Cleaning Steps



Deduplication & Nulls

Removed duplicate student records and programmatically handled blank/null values to eliminate gaps in downstream reporting.



Text Standardization

Applied Trim and Clean transformations to normalize text fields and eliminate hidden padding spaces in student names.



Range & Type Validation

Corrected data types and validated numeric boundaries across marks (0-100) and attendance percentages prior to table merging.

Data Quality Transformation

Data Attribute	Before Cleaning (Raw State)	After Power Query Cleaning	Status
Data Types	Inconsistent types (Text mixed with Numbers)	Strict explicit data type mapping per column	✔ Corrected
Duplicates	Duplicate student IDs and redundant rows	Fully deduplicated datasets	✔ Resolved
Missing Values	Unassigned blanks and null fields	Imputed and standard zeroed values	✔ Standardized
Name Formatting	Extra leading/trailing spaces in text	Cleaned and trimmed string values	✔ Validated

Consolidated Dataset Metrics



100

Total Validated Students

Academic Demographics

Span across **5 Unique Departments** covering Semesters **1 through 6**.

Core Curriculum Coverage

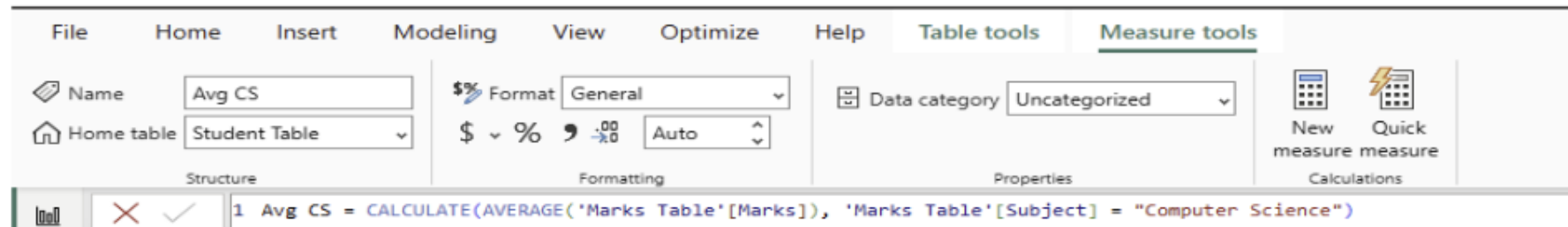
Evaluated across 3 key subjects: **Mathematics, Computer Science, and English**.

Attendance & Marks Quality

Standardized numeric percentage ranges ready for calculation of DAX KPIs.

DAX Measures Reference

#	Measure Name	DAX Expression Syntax
1	Total Students	COUNTROWS('Student Table')
2	Avg Marks Overall	AVERAGE('Marks Table'[Marks])
3	Avg Attendance	AVERAGE('Attendance Table'[Attendance %])
4	Pass %	DIVIDE([Pass Count], [Total Students], 0)
5	Subject Averages	CALCULATE(AVERAGE('Marks Table'[Marks]), 'Marks Table'[Subject] = "Mathematics")
6	Distinct Depts	DISTINCTCOUNT('Student Table'[Department])



Dashboard Core Modules



1. Student Overview

Focuses on high-level demographic distributions, total headcount KPIs, and broad cohort classification across semesters.



2. Performance Analysis

Delivers subject-level mark distributions, performance ranking algorithms, and top 10 / bottom 10 student identification.



3. Attendance Tracking

Departmental attendance heatmaps and early warning triggers to isolate at-risk students falling below thresholds.

Visualizations

Build visual





Values

Add data fields here

Drill through

Cross-report ☐

Keep all filters ☒

Add drill-through fields here

Data

Search

Attendance Table

☐ Attendance %
☐ Avg Attendance %
☐ Student ID

Marks Table

☐ Marks
☐ Maximum Marks
☐ Student ID
☐ Subject

Student Table

☐ Avg CS
☐ Avg English
☐ Avg Marks Overall
☐ Avg Marks per Student
☐ Avg Math
☐ Department
☐ Gender
☐ Name
☐ Pass %
☐ Pass Count
☐ Semester
☐ Student ID
☐ Total Students

| Key Interactive Features

-  **Dynamic Slicers:** Interactive slice-and-dice capabilities across Department and Semester dimensions.
-  **Real-Time Filtering:** Cross-filtering across visuals to allow immediate drill-downs on selected sub-cohorts.
-  **Performer Rankings:** Automated top and bottom 10 performer leaderboards for rapid academic evaluation.
-  **Low Attendance Tracking:** Automated flags highlighting students needing immediate attendance intervention.



Questions & Discussion

Ready for review and interactive dashboard demonstration.